

The Norm Check of SALSAA in folding

- 2-norm bound $\rightarrow$ inner-product $\rightarrow$ Sumcheck
- linear-time prover

Norm and inner-product

1. [2-Norm]. For a vector $\vec{v} = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, its Euclidean norm(2-norm, ℓ_2 norm) is

$$\|\vec{v}\| = \sqrt{v_1^2 + \dots + v_n^2}.$$

$$\text{So } \|\vec{v}\|^2 = v_1^2 + \dots + v_n^2 = \langle \vec{v}, \vec{v} \rangle$$

$$\text{If } \vec{v} \in \mathbb{C}^n, \|\vec{v}\|^2 = \sum_{i=1}^n v_i \bar{v}_i = \langle \vec{v}, \vec{v} \rangle$$

Norm Check (Intuition)

We will use this fact to reduce a norm-check claim to an inner-product claim, which can be further reduced to a sumcheck claim.

Step 1: Reduction to Inner-Product

Given a witness $w \in \mathbb{C}^m$, we want to prove that $\|w\|^2 \leq \nu^2$ for some bound ν . This is equivalent to proving $\langle w, \bar{w} \rangle \leq \nu^2$.

Step 2: Sumcheck Reduction

Now, we define the function $f(x)$ as $f(x) = x \cdot \bar{x}$ for $x \in \mathbb{C}$.

We want to prove that for all the entries of w ,

$$\langle w, \bar{w} \rangle = \sum_{j \in [m]} f(w_j)$$

Let $\tilde{w}(X)$ be the Multi-Linear Extension of w . Since $\tilde{w}$ agree with w at the hypercube, we have

$$\langle w, \bar{w} \rangle = \sum_{z \in \{0,1\}^\mu} f(\tilde{w}(z)) = t$$

Therefore, the verifier can check the norm bound by checking $t \leq \nu^2$, effectively reducing the norm-check to a sumcheck claim, a single evaluation of $f(\tilde{w})$ at a random point $r \in \mathbb{C}$.

Step 4: Evaluation Claims

Then, the remaining task is to prove that the evaluation $f(\tilde{w}(r)) = s$ was computed correctly.

For that, the prover sends s_0, s_1 so that:

$$s_0 = \tilde{w}(r) \quad \text{and} \quad s_1 = \bar{\tilde{w}}(r)$$

and the verifier checks that $s = s_0 \cdot s_1 \bmod q$.

By the conjugation identity $\overline{\tilde{w}(r)} = \tilde{w}(\bar{r}) \bmod q$, the claim $\tilde{w}(r) = s_1 \bmod q$ is equivalent to $\tilde{w}(\bar{r}) = \bar{s}_1 \bmod q$.

$$\overline{(a + bi)(c + di)} = \overline{(ac - bd) + (ad + bc)i} = (a - bi)(c - di) = \overline{(a + bi)(c + di)}$$

$$\overline{(a + bi)(c + di)} = (ac + bd) + (ad - bc)i = \overline{(a + bi)(c - di)} = \overline{(a + bi)(\overline{c + di})}$$

Therefore the verifier's checks amount to two evaluation claims (r, s_0) and $(\bar{r}, \bar{s}_1)$, such that:

$$s_0 = \tilde{w}(r) \quad \text{and} \quad \bar{s}_1 = \tilde{w}(\bar{r}).$$

Algebra

Cyclotomic Fields

Let $K = \mathbb{Q}(\zeta)$ be a cyclotomic field with conductor f of degree $d = \varphi(f)$

- Elements of K can be written as:

$$a_0 + a_1\zeta + a_2\zeta^2 + \cdots + a_{d-1}\zeta^{d-1}, \quad a_i \in \mathbb{Q}$$

- ζ is a root of unity of order f , ie, $\zeta^f = 1$
- φ is Euler's totient function
- $O_K = \mathbb{Z}[\zeta]$ is its ring of integers.

$$\mathbb{Z}[\zeta] = \{a_0 + a_1\zeta + a_2\zeta^2 + \cdots + a_{d-1}\zeta^{d-1} : a_i \in \mathbb{Z}\}$$

the canonical embedding $\sigma : K \rightarrow \mathbb{C}^d$. Specifically, for a given $\mathbb{Z}$ -basis $\mathbf{b} = (b_i)_{i \in [d]}$ of $\mathcal{R}$ and an element $x = \sum_{i \in [d]} x_i b_i \in \mathcal{R}$, we write $\text{cf}_{\mathbf{b}}(x) := (x_i)_{i \in [d]}$ and $\sigma(x) := (\sigma_j(x))_{j \in [d]}$, where $\sigma_j \in \text{Gal}(K/\mathbb{Q})$.

This is the group of all field automorphisms of $\mathcal{K}$ that fix $\mathbb{Q}$. For cyclotomic fields, these are exactly the maps:

$$\sigma_j : \zeta \mapsto \zeta^j, \quad \text{for } j \text{ coprime to } f$$

We extend the notation of σ naturally to vectors, i.e. if $\mathbf{x} = (x_i)_{i \in [m]} \in \mathcal{R}^m$, then $\sigma(\mathbf{x}) := (\sigma(x_i))_{i \in [m]}$ is defined as concatenations.

Norm and Trace

The ℓ_p -norm of a vector $\mathbf{x} \in \mathcal{R}^m$ is denoted by $\|\mathbf{x}\|_{\sigma,p} := \|\sigma(\mathbf{x})\|_p$. We will mostly use $\|\cdot\|_{\sigma,2}$.

For a matrix $\mathbf{M} \in \mathcal{R}^{n \times m}$, the norm is defined as $\|\mathbf{M}\|_{\sigma,p} = \max_{j \in [n]} \|\mathbf{m}_j\|_{\sigma,p}$, where $\mathbf{m}_j$ is the j -th column of the matrix.

Note that $\|\mathbf{x}\|_{\sigma,2}^2 = \text{Trace}(\langle \mathbf{x}, \bar{\mathbf{x}} \rangle)$, where $\bar{\cdot}$ denotes the complex conjugate. For a Galois extension $\mathcal{M}/\mathcal{L}$, the field trace can be computed as $\text{Trace}_{\mathcal{M}/\mathcal{L}} : K \rightarrow \mathcal{L}$,

$\text{Trace}_{\mathcal{M}/\mathcal{L}}(x) := \sum_{\sigma_j \in \text{Gal}(K/\mathcal{L})} \sigma_j(x)$. When $\mathcal{L} = \mathbb{Q}$, we write $\text{Trace} = \text{Trace}_{\mathcal{M}/\mathbb{Q}}$.

- **Trace is $\mathbb{Q}$ -linear:** For $x, y \in K$ and $a \in \mathbb{Q}$, $\text{Trace}(x + y) = \text{Trace}(x) + \text{Trace}(y)$ and $\text{Trace}(ax) = a \cdot \text{Trace}(x)$, i.e. $\text{Trace}\left(\sum_i x_i\right) = \sum_i \text{Trace}(x_i)$.

Ring Splitting

When q has multiplicative order e modulo f then $\mathcal{R}_q$ splits into d/e finite fields, i.e.

$\mathcal{R}_q \cong (\mathbb{F}_{q^e})^{d/e}$. We denote this isomorphism as $\text{CRT} : \mathcal{R}_q \rightarrow (\mathbb{F}_{q^e})^{d/e}$ and its inverse as $\text{CRT}^{-1} : (\mathbb{F}_{q^e})^{d/e} \rightarrow \mathcal{R}_q$. We naturally generalise this notation to vectors, i.e.

$\text{CRT}(\mathbf{x}) : \mathcal{R}_q^m \rightarrow (\mathbb{F}_{q^e})^{md/e}$ is defined as the concatenation of CRT applied to each entry of the vector. Furthermore, we define CRT also for polynomials, i.e. $\text{CRT} : \mathcal{R}_q^\nu[x^\mu] \rightarrow \mathbb{F}_{q^e}^{d/e}[x^\mu]$ applies CRT to each coefficient vector of the polynomial.

Norm Check Proof

This proof is to reduce a Ξ^{norm} relation to a Ξ^{lin} relation, where Ξ^{norm} is almost identical to Ξ^{lin} except that the norm of the witness is explicitly given as part of a statement:

$$\Xi_{\hat{n}, n, \mu, r, \beta}^{\text{norm}} := \left\{ \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{c}, \nu), \mathbf{W}) : \\ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{lin}} \\ \|\mathbf{W}\|_{\sigma, 2} \leq \nu \leq \beta \end{array} \right\}$$

Step 1. Norm check to sumcheck

Notice that we use the "new" definition of 2-norm, $\|\mathbf{x}\|_{\sigma, 2}^2 = \text{Trace}(\langle \mathbf{x}, \bar{\mathbf{x}} \rangle)$.

Our claim is $\langle w, \bar{w} \rangle = \sum_{z \in \{0,1\}^\mu} f(\tilde{w})(z) = t$.

So the verifier now need to check $\text{Trace}(t) \leq \nu^2$ instead of $t \leq \nu^2$ directly.

$\Pi^{\text{norm}} : \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{norm}} \rightarrow \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}}$	
$\mathcal{P}((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \nu), \mathbf{W})$	$\mathcal{V}(\mathbf{H}, \mathbf{F}, \mathbf{Y}, \nu)$
1 : parse $\mathbf{W} = (\mathbf{w}_i)_{i \in [r]}$	
2 : $\mathbf{t}^T := (\langle \mathbf{w}_i, \bar{\mathbf{w}}_i \rangle)_{i \in [r]}$	$\xrightarrow{\mathbf{t}} \mathbf{t}^T := (t_0, \dots, t_{r-1})$
3 :	for $i \in [r] : \text{Trace}(t_i) \stackrel{?}{\leq} \nu^2$
.....	
4 :	$((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{t}), \mathbf{W}) \in \Xi_{\hat{n}, n, \mu, r, \beta}^{\text{sum}}$

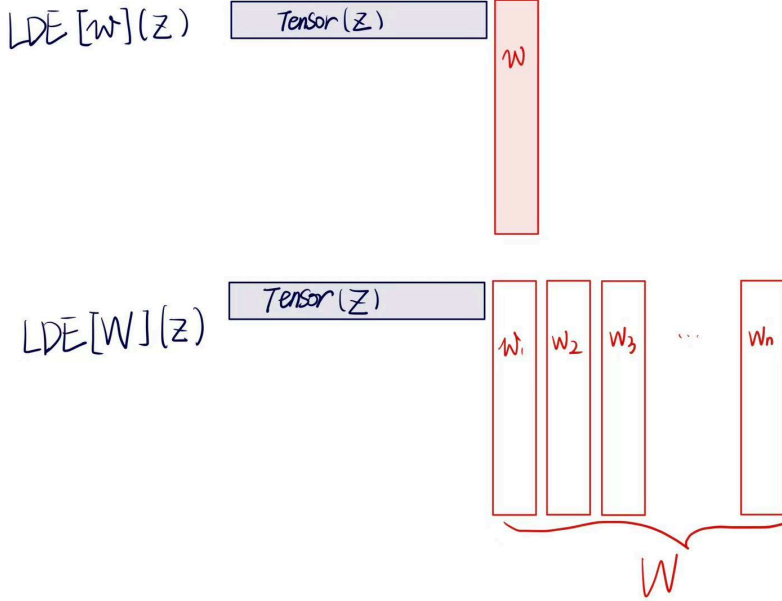
Step 2. Check $f = \tilde{w}\tilde{\tilde{w}}$

Define relation of sumcheck:

$$\Xi_{\hat{n},n,\mu,r,\beta}^{\text{sum}} := \left\{ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{t}), \mathbf{W}) : \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}} \\ \sum_{\mathbf{z} \in [d]^\mu} (\text{LDE}[\mathbf{W}] \odot \text{LDE}[\bar{\mathbf{W}}])(\mathbf{z}) = \mathbf{t} \bmod q \in \mathcal{R}_q^r \end{array} \right\},$$

where $\odot$ is the pair-wise product.

Instead of MLE, this paper applies the Low Degree Extension (LDE): the polynomial $\text{LDE}[\mathbf{W}]$ agrees with $\mathbf{W}$ at $[0, d]^\mu$, whereas the MLE agrees with $\mathbf{W}$ only at $\{0, 1\}^\mu$.



The protocol:

$\Pi^{\text{sum}}: \Xi_{\hat{n},n,\mu,r,\beta}^{\text{sum}} \rightarrow \Xi_{\hat{n},n,\mu,r,\beta,2}^{\text{lde-}\otimes}$		$\mathcal{V}((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{t}))$	
$\mathcal{P}((\mathbf{H}, \mathbf{F}, \mathbf{Y}, \mathbf{t}), \mathbf{W})$			
1 :		$u \leftarrow \mathbb{F}_{q^e}^\times$	
2 :	$\tilde{f} := \mathbf{u}^T \cdot \text{CRT}(\text{LDE}[\mathbf{W}] \odot \text{LDE}[\bar{\mathbf{W}}]) \bmod q \in \mathbb{F}_{q^e}[x^\mu]$	$\xleftarrow{\mathbf{u}^T := (u^i)_{i \in [r\varphi/e]}}$	$a_0 := \mathbf{u}^T \cdot \text{CRT}(\mathbf{t}) \bmod q$
3 :	for $j \in [\mu]$:		
4 :	$g_j(x) := \sum_{\mathbf{z}_j \in [d]^{\mu-j-1}} \tilde{f}(r_0, \dots, r_{j-1}, x, \mathbf{z}_j) \bmod q$	$\xrightarrow{g_j(x) \in \mathbb{F}_{q^e}[x]}$	$a_j \stackrel{?}{=} \sum_{z \in [d]} g_j(z) \bmod q$
5 :	<i>sumcheck for LDE</i>	$\xleftarrow{r_j}$	$r_j \leftarrow \mathbb{F}_{q^e}^\times$
			$a_{j+1} := g_j(r_j) \bmod q$
6 :	$\mathbf{r}^T := (\text{CRT}^{-1}(\mathbf{1}_{\varphi/e} \cdot r_j)^T)_{j \in [\mu]}$		
7 :	$\mathbf{s}_0 := \text{LDE}[\mathbf{W}](\mathbf{r}) \bmod q$		
8 :	$\mathbf{s}_1 := \text{LDE}[\mathbf{W}](\bar{\mathbf{r}}) \bmod q$	$\xrightarrow{(\mathbf{s}_0, \mathbf{s}_1) \in \mathcal{R}_q^{r \times 2}}$	$a_\mu \stackrel{?}{=} \mathbf{u}^T \cdot \text{CRT}(\mathbf{s}_0 \odot \bar{\mathbf{s}}_1) \bmod q$
.....			
9 :	$((\mathbf{H}, \mathbf{F}, \mathbf{Y}, (\mathbf{r}_i, \mathbf{s}_i)_{i \in [2]}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta,2}^{\text{lde-}\otimes}$ with $\mathbf{r}_0 := \mathbf{r}, \mathbf{r}_1 := \bar{\mathbf{r}}$		

$$\Xi_{\hat{n},n,\mu,\tilde{\mu},r,\beta,t}^{\text{lde}(-\otimes)} := \left\{ ((\mathbf{H}, \mathbf{F}, \mathbf{Y}, (\mathbf{r}_i, \mathbf{s}_i, \mathbf{M}_i)_{i \in [t]}), \mathbf{W}) : \begin{array}{l} ((\mathbf{H}, \mathbf{F}, \mathbf{Y}), \mathbf{W}) \in \Xi_{\hat{n},n,\mu,r,\beta}^{\text{lin}} \\ (\mathbf{r}_i, \mathbf{s}_i, \mathbf{M}_i)_{i \in [t]} \in (\mathcal{R}_q^{\tilde{\mu}} \times \mathcal{R}_q^r \times \mathcal{R}_q^{\tilde{m} \times m})^t \\ \forall i \in [t] \quad \mathbf{s}_i = \text{LDE}[\mathbf{M}_i \mathbf{W}](\mathbf{r}_i) \bmod q \end{array} \right\}$$

Comparison with Latticefold, Latticefold+

SALSAA:

- 2-norm bound
- inner-product $\rightarrow$ Sumcheck
- canonical embedding
- linear-time prover

Latticefold:

- infinite norm
- Sumcheck
- Define $g(X) := X \prod_{i \in [b]} (X - i)(X + i)$
- for all the coefficient of the witness w , $g(w_j) = 0$
- NTT and inv NTT (canonical embedding)

Latticefold+:

- infinite norm
- Sumcheck
- each entry is mapped to monomial, check the map is correct.
- additional monomial witness, double commitment